

Rafay Khan

Principal AI Engineer

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Principal AI/ML Engineer with 8+ years of experience building production AI systems across applied LLMs, computer vision, recommendation systems, and edge inference. Led end-to-end model development, evaluation, deployment, and monitoring for real-world AI applications, including multilingual customer-support assistants, high-throughput inference stacks, real-time video intelligence, and large-scale edge deployments.

PROFESSIONAL EXPERIENCE

VisionX – AI development and consulting firm building custom ML solutions

Principal AI Engineer | May 2024 – Present

Dubai, UAE / Islamabad, Pakistan

- Serving as technical lead across multiple engineering workstreams building an **LLM-powered customer support assistant** for **Iraq's largest payments processor**; supports **Arabic, Kurdish, and English** across **chat + real-time voice**, handling **7,000+ unique conversations/day** for a platform serving **30M+ end customers**.
- Delivered **~5s end-to-end response times** with autoscaling up to **2,500+ requests/sec** on **AWS ECS**, infrastructure managed via **Terraform**.
- Built production inference stack using **FastAPI, OpenAI and custom models via SGLang LLM Inference Server**, supporting **multimodal inputs** (voice, images, and text) and real-time call flows.
- Built **guardrails and evaluation workflows** for a multilingual LLM assistant, tracking latency, failure modes, escalation rates, and response quality across Arabic, Kurdish, and English interactions.
- Implemented **end-to-end observability** using **OpenTelemetry** (traces/metrics/logs) to monitor latency, failures, and debugging across distributed services.
- Defined **OpenAPI contracts** with backend teams and enabled **human-in-the-loop** escalation flows for low-confidence, high-risk, or unresolved customer interactions.

Lightly.ai – YC-backed data curation platform; Lightly Edge selects high-value data on-device

Senior Machine Learning Engineer (Consultant) | January 2024 – April 2024

Zürich, Switzerland

- Contributed to **LightlyEdge**, an on-device **smart data selection SDK** that identifies high-signal samples in real time to reduce **data transfer and storage** overhead in edge ML pipelines.
- Implemented edge inference components in **Rust** using **ONNX Runtime** to run model-driven selection logic efficiently under strict runtime constraints (Nvidia Jetson Xavier).
- Designed and integrated **model-serving primitives** (pre/post-processing, batching, interface boundaries) to support reliable inference inside a production SDK workflow.
- Contributed to **model optimization** through **quantization and pruning** workflows, helping reduce edge inference overhead while preserving predictable selection behavior under resource-constrained deployments.

Veeve.io – Smart shopping-cart retail tech using computer vision + sensor fusion

Senior Machine Learning Engineer – Team Lead | January 2020 – November 2023

Seattle, USA

- Led ML development for a **real-time in-cart video intelligence system** spanning action detection and product tracking, running at **20 FPS** across production edge deployments.
- Owned **model development, evaluation, and annotation-team workflows** for action detection and product tracking, balancing **precision/recall, latency, and edge constraints** across noisy retail environments.
- Scaled inference across **1,500 smart carts running simultaneously** across **USA, Europe, and the Middle East**, delivering stable real-time decisioning under noisy retail conditions.
- Productionized model serving on **GCP + Kubernetes**, scaling custom CV models via **NVIDIA Triton Inference Server** for high-throughput, low-latency inference.
- Built an end-to-end **ML deployment pipeline** with **GitOps-based releases** and controlled model rollouts, enabling safe promotion of model versions and faster production iteration.
- Implemented experiment tracking and reproducibility workflows with **Weights & Biases** (training runs + dataset + artifact/version tracking), supporting faster debugging and reliable model lifecycle management.
- Supported deployments with major US grocery retailers including **Albertsons** and **Kroger**, contributing to large-scale operational rollout readiness.
- Work contributed to product viability and investor outcomes, including a publicly reported **\$6.7M funding round**.

► [Watch 2-minute early prototype demo of real-time in-cart activity detection](#)

(In this video, the model identifies seven fine-grained actions including insert, remove, rummage, and theft-like behavior. Visualized via a step-function and live confidence bar chart. Since the demo, I expanded class coverage, especially for theft detection, and optimized accuracy for both edge and cloud deployment.)

Presidential Initiative for Artificial Intelligence and Computing (PIAIC)

Assistant instructor (part-time) | September 2019 – September 2020

Islamabad, Pakistan

- Selected as an **early PIAIC cohort admit (top ~1.7%)**, later serving as **Assistant Instructor** supporting learners, designing assignments and helping with technical reviews.

SPECIAL PROJECTS

[Skinsight.me](#) – Personalized AI-driven skincare product recommendation platform

Co-Founder

Dubai, UAE

- Built an AI skincare recommendation product end-to-end, spanning **frontend in Next.js**, **backend in FastAPI**, **Temporal-managed durable data ingestion pipelines** for continuous catalog growth, and an **agentic product discovery assistant** that recommends skincare products through conversational flows; all **scaled on AWS ECS**.
- Trained a **neural collaborative filtering system** on **60K+ products** and **2M+ user data points** to generate personalized recommendations and high-intent alternatives.
- Designed architecture for scalable iteration: modular pipelines, reproducible training runs, and clear interfaces between data acquisition, modeling, and product delivery.

RESEARCH EXPERIENCE

Mila

Research Fellow (part-time) | April'22 – June'24

Quebec, Canada (Remote)

Research Fellow as part of the [Fatima Fellowship](#). The primary investigator on the project **“More than more data? Comparing training with data augmentation vs IID data”**. The project aims to dissect the effect augmented samples have on representational & adversarial robustness, and model convergence in various domains.

Supervisor: Alex Hernandez-Garcia

- Designed controlled experiments comparing augmentation regimes against IID training to analyze effects on robustness, convergence, and learned representations.
- Built a modular Python experiment workbench (<https://github.com/alexhernandezgarcia/data-augmentation>) for reproducible training, evaluation, and ablation studies across progressively harder synthetic tasks.
- Developed custom augmentation methods and toy datasets to isolate how specific transformations affect model behavior and parameter learning.

TUKL-NUST R&D Lab

Research Assistant | Sep'16 – June'17

Islamabad, Pakistan

Vision research team lead for the project **“Fish Biodiversity Estimation by Low-Cost Non-Destructive Video Based Sampling (FIBEVID)”**. This project was in collaboration with Pakistan Fisheries Department, University of Kaiserslautern (TUKL) to help create a non-destructive fish bio-diversity estimation tool for Pakistan's National fish – the Mahasher. Project went on to receive funding from Hochschule RheinMain and DAAD (German Academic Exchange Service).

Supervisors: Dr. Faisal Shafiat, Dr. Imran Malik

(<https://www.hs-rm.de/de/fachbereiche/design-informatik-medien/forschungsprofil/fibevid/>)

- Created and published the first ever Mahasher fish dataset. The final dataset has 100,000 labeled images of Mahasher fish in its natural river habitat.

Research Assistant | June'16 – Sep'16 (part-time)

Summer intern at the TUKL-NUST R&D lab. Worked in collaboration with researchers at University of Kaiserslautern detect forged bank cheques under the project **“Forgery detection and segmentation of handwritten text through Hyperspectral Image Analysis.”**

Supervisor: Dr. Imran Malik

- Performed statistical analysis on hyperspectral image data to separate out inks of different varieties.

EDUCATION

National University of Sciences and Technology (NUST)

Islamabad, Pakistan

Bachelor of Science in Computer Science (BSCS)

2013 – 2017

Rector's High Achiever's List 2017 Final Year Project - Fish Biodiversity Estimation by Low-Cost Non-Destructive Video Based Sampling, FIBEVID

SKILLS

Programming Languages: Python, C/C++, Rust, Java, Javascript/Typescript, Matlab

Deep Learning / LLM Frameworks: PyTorch, TensorFlow, Keras, LangChain / LangGraph, PydanticAI

Web Frameworks: Laravel, ReactJS, D3.js, Next.js, Astro

Cloud / DevOps / MLOps: AWS, GCP, Azure, Terraform, GitHub Actions, Docker, Kubernetes, SGLang, NVIDIA Triton, ONNX Runtime, Weights & Biases

Languages: Urdu (native); English (fluent), Spanish(basic), German (basic).

SELECT HONORS & AWARDS

- **Fatima Fellowship, 2022**. Selected in top 53 fellows from an applicant pool of over 700 from 68 different countries
- **Rector's High Achievers List, 2017** final year project – FIBEVID
- **Research grant DAAD (German Academic Exchange Service), 2017** final year project – FIBEVID
- **3rd best project SEECS Open House, 2017**. Department-wide project competition at NUST – FIBEVID
- **2nd best project at COMPEEC, EME college, 2017**. Nation-wide software project competition.
- **3rd best project at NASCON'17, FAST University, 2017**. Intra-city software project competition.

SELECT BLOGS

- [Nothing but NumPy: Understanding & Creating Neural Networks with Computational Graphs from Scratch](#)
 - 50K+ views; KDnuggets Gold Award
- [Nothing but NumPy: Understanding & Creating Binary Classification Neural Networks with Computational Graphs from Scratch](#)
 - 48K+ views; Published in Towards Data Science
- [Where did the Binary Cross-Entropy Loss Function come from?](#)
- [Why Using Mean Squared Error \(MSE\) Cost Function for Binary Classification is a Bad Idea?](#)
- [How do TensorFlow and Keras implement Binary Classification and the Binary Cross-Entropy function?](#)